RAMYA RAGU

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LinkedIn | GitHub | Portfolio

EDUCATION

- **Vrije Universiteit** *2024 - 2026*
  MASTER OF SCIENCE IN ARTIFICIAL INTELLIGENCE - HEALTHCARE SPECIALIZATION Amsterdam, Netherlands
  - CGPA: 7.9/10 *(Dutch scale)*
  - Thesis Grade: 9/10

- **SRM Institute of Science and Technology** *2020 - 2024*
  BACHELOR OF TECHNOLOGY, COMPUTER SCIENCE AND ENGINEERING - AI & ML SPECIALIZATION Chennai, India
  - CGPA: 9.59/10
  - Thesis Grade: 10/10

PATENTS AND PUBLICATIONS

C = CONFERENCE, P = PATENT APPLICATION, T = THESIS, U = UNDER REVIEW

[U.1] Elena Beretta, Ramya Ragu (2026). **When Clinical Predictions Become Narratives: A Risk Communication Contract for LLM-Generated Survival Explanations**, September 2026, *ACM CHI 2027* (under review)

[T.3] Ramya Ragu. (2026). **Evaluation of LLM-Generated Survival Explanations: Integrating Prompt Design, Confidence, Stability, and Grounding in Synthetic Clinical Narratives**, July 2026, Vrije Universiteit, Amsterdam, Netherlands.

[P.1,T.2] Paavai Anand, Ramya Ragu, Srinivas Muralidharan. (2024). **A System And Method For Information Extraction And Knowledge Organization In Biomedical Literature**. Indian patent application no. 202441093281 (under examination, First Examination Report issued)

[T.1, C.2] Ramya Ragu, Niveditha S (2023). **Developing a Personalized Clinical Support System Utilizing Classification Models for Drug Assessment**. In *The National Conference on Technology for the Society (NCTS'23)*, October 2023, SRMIST, Chennai, Tamil Nadu, India. *Awarded Best Paper - Session Recognition.*

[C.1] Ramya Ragu. (2022). **Clinical Trials Selection for Cancer Patients using NLP**. Paper presented at *UBERTECH'22*, October 2022, SRMIST, Chennai. *Awarded Second Place*

RESEARCH & PROJECTS

- **LLM Confidence Calibration for Clinical Survival Explanations** *Amsterdam, Netherlands*
  MASTER'S THESIS - Vrije Universiteit Amsterdam JAN 2026 - JULY 2026
  - Evaluated LLM behaviour across eight prompt and decoding configurations for pancreatic cancer survival risk communication (MedGemma, Ollama).
  - Built an automated evaluation framework scoring numerical correctness, risk-label agreement, robustness, confidence calibration, and narrative faithfulness across 400 explanations for 50 synthetic patients, comparing self-reported confidence with internal signals (token log-probabilities and entropy).
  - Observed that confidence-signal credibility depends on prompt style and that misplaced certainty poses a greater clinical safety risk than hallucination, suggesting signal-specific calibration requirements before deployment.
  - *Extended into a paper submitted to ACM CHI 2027 (under review). Code available on request.*

- **LLM-Based Conversational Agent for Psychiatric Stress Analysis** *Amsterdam, Netherlands*
  RESEARCH INTERN - Amsterdam UMC, Dept. of Psychiatry, Stress in Action JUNE 2025 - JULY 2025
  - Designed prompt strategies and applied local LLMs (Ollama) for generating stress-level conversations across clinical timelines.
  - Defined model selection and pipeline architecture for the agent's early-stage development, working with psychiatric researchers on clinical requirements.

- **Hybrid LLM-Ontology Extraction for Knowledge Graphs** *Amsterdam, Netherlands*
  COURSE PROJECT - Vrije Universiteit Amsterdam (Knowledge Graphs & Semantics) FEB 2025 - MARCH 2025
  - Built a hybrid pipeline combining LLM-based entity and relation extraction with existing domain ontologies, refined through similarity metrics and confidence scoring.
  - Compared manual, automated, and hybrid approaches for entity and relation extraction. The hybrid method substantially reduced hallucinations, improved semantic consistency, and scaled better across large corpora.

- **MedOntoKG: Ontology-Enhanced Biomedical Knowledge Extraction** *Chennai, India*
  BACHELOR'S THESIS - SRM Institute of Science and Technology NOV 2023 - MAY 2024
  - Developed an ontology-driven knowledge graph that structures biomedical abstracts on neurodegenerative diseases into diagnosis, prognosis, and treatment entities (Neo4j, Protégé).
  - Fine-tuned XLNet for named entity recognition across ∼40 biomedical entity types (weighted F1 0.69) and for abstract classification into diagnosis, prognosis, and treatment (85% accuracy, weighted F1 0.85), then inferred relations between extracted terms via cosine similarity of the fine-tuned embeddings.
  - Resulted in Indian Patent Application No. 202441093281 (published; awaiting examination).

Professional Experience

- **Teaching Assistant - Vrije Universiteit Amsterdam** *SEPTEMBER 2025 - MAY 2026*
  Amsterdam, Netherlands
  - **Natural Language Processing** (Apr-May 2026): Led lab sessions on NLP pipelines, transformers and large language models (NLTK, spaCy, Hugging Face). Assessed student projects on implementation and experimental design across basic and advanced NLP.
  - **Evolutionary Computing** (Sep-Oct 2025): Supported students implementing genetic algorithms and evolution strategies for neuroevolution, including fitness-function design and hyperparameter tuning. Assessed projects on algorithmic efficiency and experimental design on Ariel Space Telescope mission sub-tasks.
- **Junior Data Analyst - SRM Institute of Medical Sciences (SIMS)** *FEBRUARY 2023 - APRIL 2023*
  Chennai, India
  - Built an interactive Streamlit dashboard tracking patient behaviour, revenue trends and expenses, and a revenue-analysis tool supporting resource-allocation decisions.

Skills

Programming Languages	Python, SQL, Cypher
Tools	Git, LaTeX, Ollama
Software	Neo4j, Protégé, Microsoft Azure
Frameworks & Libraries	spaCy, NLTK, Hugging Face Transformers, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Streamlit, RDFLib

Certification Courses

- **AI for Medicine - DeepLearning.AI** (Link to Certificate)
- **Master Deep Learning Using Case Studies: Beginner-Advance - Geekshub Pvt Ltd** (Link to Certificate)
- **Microsoft Certified: Azure AI Fundamentals - Microsoft** (Link to Certificate)
- **Skin Cancer Imaging Challenge 2025 - Medical AI - VUMC** (Link to Certificate)

Additional Information

Languages:

- English (Proficient)
- Tamil (Native)
- Dutch (Beginner)

Extracurricular Activities:

- Volunteered at Ubertech'22 and Technozare'22 tech fests (SRMIST), incl. a workshop on object detection and tracking in surveillance videos.
- Bharatanatyam dancer - performed at SRMIST and VU Amsterdam cultural events; active member of NL_Adavumeet, leading practice sessions
- Board game enthusiast - interested in how people reason about strategy and choices.